

Technology News

Robot capable of learning

A humanoid robot about the size of a three-year-old child is learning to think for itself. It's part of the RobotCub project that aims to study how humans learn and think by using a robot with the size and brain of a toddler. Called iCub, the robot is designed to learn from experience and adapt to changes in its environment, just like a human child.

Human intelligence develops through interaction with the environment and other human beings, and mental processes are strongly associated with the physical body and its actions. The central hypothesis of the project is therefore that the best way to model the human mind is to create a humanoid that is controlled by realistic algorithms and allowed to explore the world like a real child.

The big eyes of iCub robot allow it to see and track objects in its environment. Many electronic circuits built into the articulated trunk and limbs give it a wide range of movements.

Sensors allow the robot to feel.

The iCub was developed at the Italian Institute of Technology (ITT) in Genoa, Italy, where the RobotCub project began in 2004 under the leadership of Giulio Sandini.

Robot controlled by human brain cells

A robot controlled by human brain cells could soon become a reality. Kevin Warwick and Ben Whalley at the University of Reading, the UK, have already used rat brain cells to control a simple wheeled robot.

Some 300,000 rat neurons grown in a nutrient broth and producing spikes of electrical activity were connected to the output of the robot's distance sensors. The neurons proved capable of steering the robot around a small enclosure.

The researchers say that observing how their neuron culture responds to stimulation could improve understanding of neurological conditions such as epilepsy. For instance, the way large

numbers of neurons sometimes spike in unison—a phenomenon known as 'bursting'—may be similar to what happens during an epileptic seizure. If that behaviour can be altered by changing the culture chemically, electrically or physically, it might hint at potential therapies.

To make the system a better model of human disease, a culture of human neurons will be connected to the robot once the current work with rat cells is completed. This will be the first instance of human cells being used to control a robot.

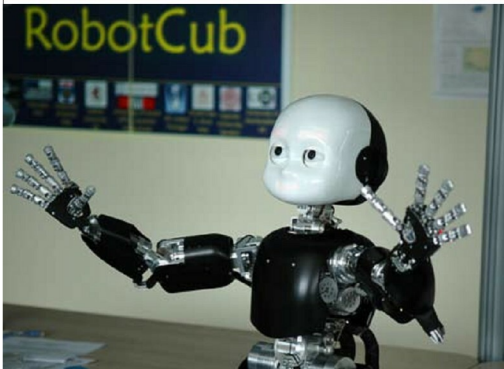
Bio-nano device to better electronics

Mingling biological components in electronic circuits could enhance the efficiency of biosensing and diagnostic tools, neural prosthetics such as cochlear implants, and even future computers.

Researchers at Lawrence Livermore National Laboratory (LLNL) have already devised a versatile hybrid platform that uses lipid-coated nanowires to build prototype bionanoelectronic devices.

To create the bionanoelectronic platform, the LLNL team turned to lipid membranes, which are ubiquitous in biological cells. These membranes form a stable, self-healing and virtually impenetrable barrier to ions and small molecules. They can also house an unlimited number of protein machines that perform a large number of critical recognition, transport and signal transduction functions in the cell.

The researchers incorporated lipid bilayer membranes into silicon nanowire transistors by covering the nanowire with a continuous lipid bilayer shell that forms a barrier between the nanowire surface and solution species. This 'shielded



iCub robot with the size and brain of a three-year-old child learns from experience